

The University Interscholastic League

Number Sense Test • HS B • 2026

Final	_____
2nd	_____
1st	_____
Score	Initials

Contestant's Number _____

**Read directions carefully
before beginning test**

**DO NOT UNFOLD THIS SHEET
UNTIL TOLD TO BEGIN**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. ALL PROBLEMS ARE TO BE SOLVED MENTALLY. Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

STOP -- WAIT FOR SIGNAL!

- | | |
|---|--|
| <p>(1) $213 + 314 + 2026 =$ _____</p> <p>(2) $202.6 - 21.3 - 31.4 =$ _____ (decimal)</p> <p>(3) $\frac{2}{13} \div \frac{3}{14} =$ _____</p> <p>(4) $2\frac{1}{3} \times 3\frac{1}{4} =$ _____ (mixed number)</p> <p>(5) Which is smaller $\frac{2}{13}$ or $\frac{3}{14}$? _____</p> <p>(6) $[(26 - 13) + (26 - 12)]^2 =$ _____</p> <p>(7) The GCD of 78 and 51 is _____</p> <p>(8) $20 \times 13 + 20 \times 14 =$ _____</p> <p>(9) $4 + 7 + 10 + \dots + 31 + 34 =$ _____</p> <p>*(10) $21326 + 31426 - 2026 =$ _____</p> <p>(11) $333 \times \frac{3}{37} =$ _____</p> <p>(12) Given: {2, 1, 3, 2, 6, 3, 1, 4, 2, 6}. Find the sum of the median and the mode. _____</p> <p>(13) The sum of the positive divisors of 26 is _____</p> <p>(14) 1314×15 is _____</p> <p>(15) $1495 \times 5 - 25 =$ _____</p> <p>(16) $17 \times \frac{17}{21} =$ _____ (mixed number)</p> <p>(17) $21^2 - 17^2 = 38 \times$ _____</p> | <p>(18) $0.75\% =$ _____ (fraction)</p> <p>(19) 2 dekagram = .7 oz. and 21 oz. = _____ grams</p> <p>*(20) $21.3 \times 31.4 + 2026 =$ _____</p> <p>(21) $3\frac{1}{4} \times 8\frac{2}{3} =$ _____ (mixed number)</p> <p>(22) $\{2, 13, 20, 26\} \cup \{3, 14, 20, 26\}$ contains how many distinct elements? _____</p> <p>(23) $\sqrt[3]{2744} =$ _____</p> <p>(24) 21% of 34 is _____ % of 238</p> <p>(25) $\frac{9}{16} \div \frac{9}{64} \times \frac{9}{256} =$ _____</p> <p>(26) $2x - 13 = 3x - 14$ and $x =$ _____</p> <p>(27) The LCM of 78 and 51 is $78 \times$ _____</p> <p>(28) 75 is 41.666...% of _____</p> <p>(29) The sum of the prime numbers greater than 21 and less than 31 is _____</p> <p>*(30) $\sqrt{314213} =$ _____</p> <p>(31) The 14th term of the sequence 5, 10, 15, 20, ... is 70. The 13th term is _____</p> <p>(32) The additive inverse of 0.41666... is _____</p> <p>(33) How many 4-digit numbers can be made using these five the digits 3, 1, 4, 2 and 6? _____</p> |
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- (34) If $\frac{13x-2}{13x+2} + \frac{13x+2}{13x-2} = 2 + \frac{B}{169x^2-4}$, then $B =$ _____
- (35) $46 \times 46 =$ _____
- (36) $314 =$ _____₆
- (37) The area of a rectangle with length 13 cm is 182 cm². Find its perimeter. _____ cm
- (38) $\frac{3}{4} - \frac{16}{19} =$ _____
- (39) $(13^3 + 14^3 - 26) \div 27$ has a remainder of _____
- *(40) A half mile is _____ centimeters
- (41) $22 \times 82 =$ _____
- (42) $\{2, 13, 20, 26\} \cap \{3, 14, 20, 26\}$ contains how many distinct elements? _____
- (43) $\frac{23}{33} = 0.ababab\dots$ and $b - a =$ _____
- (44) If $\frac{1}{3} + \frac{1}{6} + \frac{1}{10} + \dots + \frac{1}{n} = \frac{7}{9}$, then $n =$ _____
- (45) The hypotenuse of a 45° - 45° - 90° triangle is $\sqrt{50}$. The lengths of the other sides total _____
- (46) Let $(2x+3)(3x-4) = ax^2 + bx + c$. Find $a + b + c =$ _____
- (47) If ${}_7P_2 = k \times ({}_7C_5)$, then $k =$ _____
- (48) $(4+3i)(2+i) = a+bi$ and $a+b =$ _____
- (49) Let $(-3, 4)$ be the midpoint of the segment with endpoints $(2, 6)$ and (x, y) . Find $x+y$. _____
- *(50) $5202 \times (213 - 314) =$ _____
- (51) $2026_8 - (213_8 + 314_8) =$ _____₈
- (52) How many 4-digit numbers less than 4000 can be made using these four digits 1, 2, 3, and 4? _____
- (53) The sum of the digits of a 3-digit number is 13. How many such numbers exist? _____
- (54) A sock drawer contains 8 white and 5 brown socks. If the first sock drawn is brown what is the probability the second one is brown? _____
- (55) $21331_6 \div 4_6$ has a remainder of _____₆
- (56) $2x^3 + 3x^2 - 3x - 2 = 0$ and P, Q, and R are the roots. Find $PQR - (PQ + PR + QR)$. _____
- (57) $\prod_{k=2}^{k=6} (k-1) =$ _____
- (58) The point $(3, -4)$ is reflected across the line $y = -x + 1$ to the point (h, k) . Find $h+k$. _____
- (59) $5^3 - 2 =$ _____₅
- *(60) $\sqrt[3]{21331426} =$ _____
- (61) $2x - y = 3$ and $3x - y = 4$. Find $x =$ _____
- (62) $2 + 1 \div 3 - 3! \times (1 - 4) \times 2 \div 6 =$ _____
- (63) $\begin{vmatrix} 2 & 13 \\ 3 & 14 \end{vmatrix} =$ _____
- (64) $43^{21} \div 22$ has a remainder of _____
- (65) If $0.1444\dots$ base 8 = $\frac{a}{b}$ base 8, then $a+b =$ _____
- (66) The y-intercept of the line through points $(2, 13)$ and $(3, 14)$ is at $y =$ _____
- (67) If $\sec \theta = -4$, $\theta \in \text{QII}$, then $\cos \theta =$ _____
- (68) $\lceil \sqrt{10} - \sqrt{5} \rceil =$ _____
- (69) $0.34_6 =$ _____ (fraction)
- *(70) The volume of a rectangular prism with length 13 m, width 14 m, and height 27 m is _____ m³
- (71) If $xy = 3$ and $x - y = 14$ then $x^3 - y^3 =$ _____
- (72) $2133 \times 14 =$ _____
- (73) The roots of $2x^3 + 3x^2 - 3x - 2 = 0$ are d, e, and f. Find $(d+e)(e+f)(f+d)$. _____
- (74) Find the slope of the tangent to $f(x) = 3x^2 - 3x - 4$ at $x = 2$. _____
- (75) Let $f(x) = \frac{2x-13}{3}$. Find $f^{-1}(14)$. _____
- (76) Let $f(x) = \frac{2x+13}{14-3x}$. The horizontal asymptote of $f(x)$ is at $y =$ _____
- (77) $g(x) = 2x^3 + 13x^2 - 3x - 14$ and $g'(-1) =$ _____
- (78) $\int_0^3 \int_3^4 (x-y) dx dy =$ _____
- (79) $203 \times 304 =$ _____
- *(80) $(21303142026)^{\frac{1}{4}} =$ _____

University Interscholastic League - Number Sense Answer Key HS • Invitation B • 2026

*number) $x - y$ means an integer between x and y inclusive

NOTE: If an answer is of the type like $\frac{2}{3}$ it cannot be written as a repeating decimal

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|---------------------------------------|----------------------|--------------------------------|--|
| (1) 2,553 | (18) $\frac{3}{400}$ | (34) 16 | (57) 120 |
| (2) 149.9 | (19) 600 | (35) 2,116 | (58) 3 |
| (3) $\frac{28}{39}$ | *(20) 2,561 — 2,829 | (36) 1242 | (59) 443 |
| (4) $7\frac{7}{12}$ | (21) $28\frac{1}{6}$ | (37) 54 | *(60) 264 — 291 |
| (5) $\frac{2}{13}$ | (22) 6 | (38) $-\frac{7}{76}$ | (61) 1 |
| (6) 729 | (23) 14 | (39) 1 | (62) $\frac{25}{3}, 8\frac{1}{3}$ |
| (7) 3 | (24) 3 | *(40) 76,444 — 84,490 | (63) — 11 |
| (8) 540 | (25) $\frac{9}{64}$ | (41) 1,804 | (64) 21 |
| (9) 209 | (26) 1 | (42) 2 | (65) 83 |
| *(10) 48,190 — 53,262 | (27) 17 | (43) 3 | (66) 11 |
| (11) 27 | (28) 180 | (44) 36 | (67) — .25, $-\frac{1}{4}$ |
| (12) 4.5, $\frac{9}{2}, 4\frac{1}{2}$ | (29) 52 | (45) 10 | (68) 1 |
| (13) 42 | *(30) 533 — 588 | (46) — 5 | (69) $\frac{11}{18}$ |
| (14) 19,710 | (31) 65 | (47) 2 | *(70) 4,669 — 5,159 |
| (15) 7,450 | (32) $-\frac{5}{12}$ | (48) 15 | (71) 2,870 |
| (16) $13\frac{16}{21}$ | (33) 120 | (49) — 6 | (72) 29,862 |
| (17) 4 | | *(50) — 551,673 —
— 499,131 | (73) 1.25, $\frac{5}{4}, 1\frac{1}{4}$ |
| | | (51) 1277 | (74) 9 |
| | | (52) 18 | (75) 27.5, $\frac{55}{2}, 27\frac{1}{2}$ |
| | | (53) 69 | (76) $-\frac{2}{3}$ |
| | | (54) $\frac{1}{3}$ | (77) — 23 |
| | | (55) 3 | (78) 6 |
| | | (56) — .5, $-\frac{1}{2}$ | (79) 61,712 |
| | | | *(80) 363 — 401 |